H. Additional information

In this section we expand on linear factorization, make a note on the non-triviality of linear factorization, and expand on the reasoning of using whitening in measuring linear factorization. In Section H.1 we summarize the overall procedure of measuring linear factorization. In Section H.3 we provide an intuition of linear factorization through a simple example. In Section H.4 we show that linear factorization is not a trivial property of linearly compositional models, and illustrate a few cases where the representation cannot be decomposed into a sum of per-concept components even under perfect classification.

H.1. Testing linear factorization

Large pre-trained models may encode information beyond the specific concepts in our dataset. To isolate the conceptual structure, we train per-concept linear probes. For each concept $i \in [k]$ and value j, we learn a linear probe $\boldsymbol{w}_{i,j}$, form the probe matrix $\boldsymbol{W} \in \mathbb{R}^{m \times d}$, where m is the number of values across all concepts, and project embeddings onto the joint probe span. We do this by first computing the projection matrix $P_{\boldsymbol{W}}$ and then projecting the embeddings onto the joint probe span.

We report *Projected* R^2 after projecting embeddings onto the probe span. To prevent trivial high scores from dominant directions, we whiten the embeddings by applying PCA and normalizing to unit covariance. We compute metrics on $P_{\boldsymbol{W}}\boldsymbol{z}$ after PCA-whitening, applying the same transform to data and reconstructions. We elaborate on this below.

H.2. Whitening in measuring linear factorization

We need to be cautious when assessing the degree of linearity in the representations, otherwise, we may mistake high R^2 scores for linear factorization when in fact the representation is not linearly factored. For example, if certain concept values dominate the variance in the representation, the R^2 may be inflated. To address this, in the main experiments in Section 5.1 we whiten the representations by applying PCA and normalizing to unit covariance. This ensures that a few dominant directions do not dominate the variance in the representation. If the representations are already linearly factored, this will not affect the R^2 score.

We illustrate this through three examples in a hypothetical two-dimensional representation space with two concepts in Fig. 19. In the first case ((**a**)) the representation is already linearly factored: each embedding is written as a sum of two concept components without noise. This yields an R^2 score of 1; whitening does not change the score.

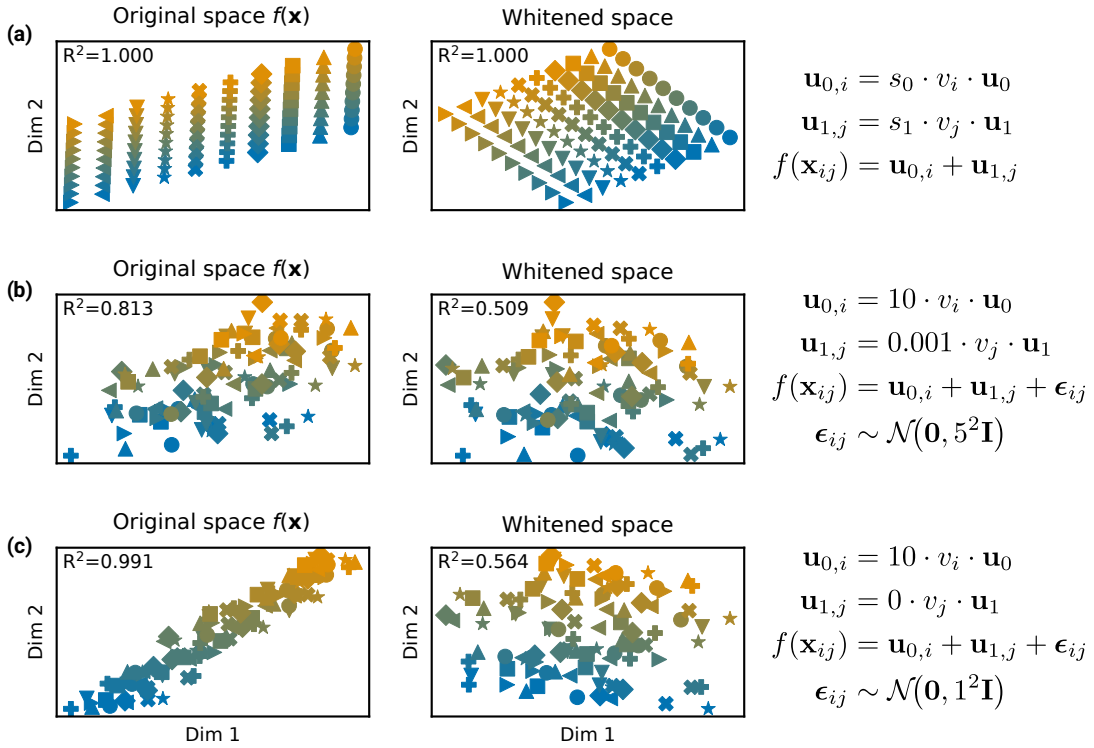

Figure 19. **Whitening in measuring linear factorization**. The representation is not linearly factored, but the R^2 score is high due to the dominance of the dominant direction.

In the second case ((**b**)) the representation is partly linear, but the noise $\boldsymbol{\epsilon}_{ij}$, independent of the concept values, dominates the overall

variance. Since the scale of the noise is generally lower than the scale of the first concept component, the R^2 score is high at 0.813. Whitening, however, removes the dominant direction, and the R^2 score drops to 0.509.

Lastly, in the third case ((**k**)) the representation does not express any information about the second concept, yet the R^2 score is still high at 0.991. Again, whitening reveals the underlying issue and changes the score to 0.564 due to the noise in the embeddings.

H.3. Intuition of linear factorization

We measure the extent of linearity present in the embeddings through the R^2 score. Intuitively, the score quantifies how well the representation can be decomposed into a sum of per-concept components. Recall from Definition 1 that we assume a presence of k concepts, each of which can take any of the n values. A value of $R^2 = 1$ indicates that the representation can be perfectly decomposed into a sum of per-concept components.

We illustrate a few examples to give intuition. We consider a two-dimensional representation space with two concepts ($k = 2$). In the first case, we consider a case of 24 values per concept ($n = 24$). In the second case, we consider a case of 6 values per concept ($n = 6$). In both cases the reported R^2 are w.r.t. the whitened space.

The first case (Fig. 20, (**a**)) exhibits perfect linearity in the embeddings with $R^2 = 1$. In this case, the $n^2 = 24^2 = 576$ can be perfectly generated using only $2 \cdot 24 = 48$ vectors in $\mathbb{R}^2$. The second and third columns of the plot show the approximations of the underlying factors $\boldsymbol{u}_{0,i}, \boldsymbol{u}_{1,j}, i, j \in [n]$. As expected, using these approximate factors allows us to perfectly reconstruct the representation, shown in the fourth column.

The second case (Fig. 20, (**b**)) exhibits lower degree of linearity with $R^2 = 0.53$. As such, we cannot perfectly reconstruct the representation using only the approximate factors, as shown in the last column of the plot.

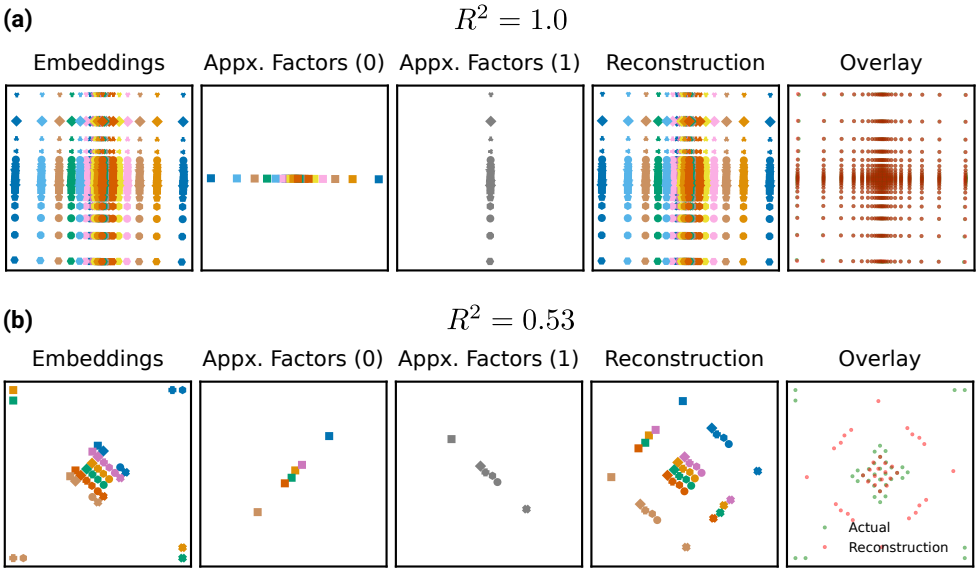

Figure 20. **Intuition of linear factorization**. In (**a**) the representations can be perfectly reconstructed by a set of per-concept components, while in (**b**) they are insufficient to reconstruct the representation. Refer to the text for more details.

H.4. Non-triviality of linear factorization of linearly compositional models

Recall that linearly compositional models (though not necessarily generalizable ones), as defined in Definition 3, admit a set of probes that can perfectly classify all inputs in the grid $\mathcal{C}$. Proposition 1 shows that linearly compositional models must exhibit linear factorization. This naturally raises the converse question: does the mere existence of a set of perfect linear classifiers imply linear factorization? We answer in the negative.

The intuition is as follows. As per Desideratum 1, linearly compositional models need to divide the representation space into all possible combinations of concept values, n^k of them. Each region within the n^k partitions must contain the corresponding combination of concept values. Under linear factorization, the degrees of freedom of the embeddings within each cell are low, yielding an R^2 score of 1. However, even if linear factorization initially holds, the embeddings can generally be perturbed to violate the linear factorization constraint while still being contained within the correct cell.

To illustrate this point, we consider two general cases: (i) the number of concepts is equal to the dimension of the embeddings ($k = d$), and (ii) the number of concepts is less than the dimension of the embeddings ($k < d$). As detailed in Section 4.3, case (i) is tight (the dimension

cannot be further reduced), while case (ii) is not. In both cases we assume two concepts and an embedding space that admits two linear probes, one for each concept. Additionally, in both cases we illustrate separately the argmax regions where a certain concept value is predicted ($\mathcal{R}_{i,j}, i \in [2], j \in [n]$), and the region where a certain combination of concept values is predicted ($\mathcal{R}_{0,j} \cap \mathcal{R}_{1,k}, j, k \in [n]$, as per Desideratum 1).

The first concept values' regions in the embedding space are shown in blue, while the second concept values' regions are shown in orange.

Case (i): $k = d$**.** In Fig. 21, **(a), (b)** we show two cases that exhibit perfect linear classification. In **(a)** a few outliers violate the linearity of the representation, which is also reflected in the $R^2 = 0.53 < 1$. In **(b)** the argmax regions are highly irregular, and the majority of the embeddings are almost intersecting the decision boundaries, resembling an extremely brittle embedding space susceptible to adversarial attacks, though the classification accuracy is still 100%.

Case (ii): $k < d$**.** In Fig. 21, **(k), (d), (e)** we show three cases that exhibit perfect linear classification, but with linearity scores ranging from $R^2 = 0.32$ to $R^2 = 0.83$. Because of the higher degrees of freedom, the embeddings enjoy even more space to be perturbed while still exhibiting perfect linear classification.

Overall, these points illustrate that linear factorization is not a trivial property of linearly compositional models, even when perfect classification holds.